MANIPULATED IMAGE DETECTION AND SEGMENTATION

Deep Learning Semester Project Proposal

BSCS VII

Student 1: Karan Kumar	**CMS: 023-22-0122**
Student 2: Behzad Hassan	**CMS: 053-22-0008**
Student 3: Ubedullah	**CMS: 023-22-0081**

Instructor:
Dr. Sher Muhammad Daudpota

SUKKUR IBA UNIVERSITY
Computer Science Department

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Table of Contents

List of Figures

Abstract

Digital image manipulation has become a growing concern in the modern era of social media, journalism, and legal evidence handling [1][2]. With the advancement of generative AI and powerful editing tools, forged images can be created and shared effortlessly, raising questions about visual authenticity and digital trust [3][4]. This project proposes a deep learning–based approach for detecting and localizing manipulated regions in digital images using the DEFACTO Image Forgery Dataset [5].

The proposed system consists of two main components: a Convolutional Neural Network (CNN) for classifying images as real or forged [6], and a U-Net or DeepLabV3+ model for segmenting and visualizing the manipulated regions [7][8]. The dataset will be preprocessed through resizing, normalization, and augmentation [9], followed by model training using optimized loss functions such as Binary Cross-Entropy and Dice Loss [10]. Performance evaluation will be conducted using Accuracy, Precision, Recall, F1-Score, Intersection over Union (IoU), and Dice Coefficient metrics [11][12].

The expected outcome of this study is a robust model capable of achieving high accuracy in both detection and localization, with clear visualization of tampered areas [13]. By combining classification and segmentation within a unified framework, this project aims to contribute to digital forensics, image integrity verification, and media authenticity through explainable deep learning techniques [14][15].

1.0 Introduction

In today's digital era, image manipulation has become increasingly sophisticated due to the rapid advancement of image editing tools and generative AI technologies [1][2]. While such tools enable creativity and artistic expression, they also pose serious threats to digital integrity, online authenticity, and media credibility. Images altered to mislead or deceive are frequently used in misinformation campaigns, legal disputes, and digital forensics, making forgery detection a crucial research area in computer vision [3][4].

Traditional image forgery detection methods, based on handcrafted features such as noise inconsistencies or JPEG artifacts, have shown limited success in identifying complex manipulations [5]. With the emergence of Deep Learning, Convolutional Neural Networks (CNNs) have demonstrated remarkable capability in automatically learning discriminative spatial features from large-scale image data, enabling higher accuracy in classifying manipulated images [6][7]. However, while most existing methods can detect whether an image has been tampered with, localizing the exact manipulated regions remains a major challenge [8].

To address this limitation, this project proposes a deep learning–based system that can both detect and localize image forgeries using the DEFACTO Dataset, a comprehensive benchmark for image manipulation detection [9]. The proposed approach integrates a CNN-based classification model to determine whether an image is authentic or forged [6], and a U-Net (or DeepLabV3+) based segmentation model to generate precise binary masks highlighting tampered regions [10][11]. By combining detection and localization in a unified workflow, this study aims to contribute to the field of digital forensics and image authenticity verification with improved accuracy and interpretability [12][13].

2.0 Problem Identification

2.1 Background

With the exponential growth of digital content, image manipulation has emerged as a serious concern across social media, journalism, law enforcement, and digital forensics [1][3]. Modern editing software such as Adobe Photoshop, GIMP, and AI-based generative models (e.g., GANs, diffusion models) can produce highly realistic forgeries that are nearly indistinguishable from authentic images to the human eye [2][4].

Manipulated images are widely used to spread misinformation, fabricate evidence, or alter public perception, posing threats to both individual reputation and societal trust in visual media [3][4].

Traditional forgery detection methods rely on handcrafted features such as color inconsistencies, noise patterns, or JPEG compression artifacts [5]. However, these methods often fail to detect complex and high-quality manipulations, particularly when forgeries involve semantic-level editing such as object removal, content swapping, or AI-generated enhancements [4].

With the advent of Deep Learning, particularly Convolutional Neural Networks (CNNs), automatic feature learning has enabled better performance in distinguishing between real and manipulated images [6][7]. Yet, most existing models still operate as binary classifiers - they can identify whether an image is fake but cannot localize or visualize the tampered areas [8].

This lack of spatial interpretability limits their practical utility in forensic and investigative applications where pinpointing the manipulated region is essential [12][13].

2.2 Research Questions

The core questions guiding this project are:

1. Detection – How can a deep learning model be designed to accurately distinguish between authentic and manipulated images using the DEFACTO dataset?
2. Localization – Can a segmentation network (e.g., U-Net or DeepLabV3+) effectively identify and highlight the precise regions that have been manipulated?
3. Performance & Generalization – How well does the proposed dual-model approach perform across various types of manipulations (e.g., copy-move, splicing, object removal, morphing), and can it generalize to unseen forgery types?
4. Interpretability – Can the integration of classification and segmentation improve explainability in image forgery detection systems?

2.3 Hypothesis

It is hypothesized that:

1. A CNN-based classification model can learn discriminative features that differentiate authentic and manipulated images with high accuracy.
2. A segmentation model such as U-Net or DeepLabV3+, trained with Binary Cross-Entropy (BCE) and Dice Loss, can accurately localize manipulated regions by generating precise binary masks.
3. Combining both models within a unified detection–localization pipeline will enhance both accuracy and interpretability, outperforming standalone classification or segmentation approaches.
4. Through effective preprocessing, augmentation, and training strategies, the proposed system will generalize well to diverse forgery types within the DEFACTO dataset.

2.4 Contributions of the Project

This project aims to make the following contributions to the field of image forensics and deep learning–based forgery detection:

1. Dual-Stage Detection Framework:

Development of a two-stage deep learning pipeline that integrates image-level classification and pixel-level segmentation for comprehensive forgery analysis.

2. Localization of Manipulated Regions:

Implementation of a segmentation network capable of visualizing tampered regions, providing spatial interpretability that enhances forensic trust and usability.

3. Use of a Diverse Benchmark Dataset (DEFACTO):

Leveraging the DEFACTO dataset, which includes multiple manipulation types and ground-truth masks, to train and evaluate models with diverse and realistic examples.

4. Comprehensive Evaluation:

Detailed analysis using multiple metrics, Accuracy, Precision, Recall, F1-Score, IoU, Dice Coefficient - to ensure balanced and reliable assessment of both detection and localization performance.

5. Practical Application for Digital Forensics:

Providing a prototype that can serve as a foundational system for media verification tools, authenticity checks, and forensic analysis, contributing to digital trust and misinformation mitigation.

3.0 Objectives

The primary objectives of this project are as follows:

1. To design a classification model that can distinguish between authentic and forged images with high accuracy.

2. To implement a segmentation network (such as U-Net or DeepLabV3+) to generate precise binary masks that highlight manipulated areas.
3. To evaluate and compare the model's performance using quantitative metrics such as Accuracy, Precision, Recall, F1-Score, IoU, and Dice Coefficient.
4. To analyze qualitative results through visual comparison of predicted masks and ground truth masks for different forgery types.

4.0 Dataset Discussion

The proposed system will utilize the DEFACTO (Deepfake and Image Forgery Detection) Dataset, which is a large-scale benchmark dataset specifically designed for training and evaluating deep learning models in image manipulation detection and localization tasks [16]. The DEFACTO dataset includes a diverse collection of images representing various types of manipulations, such as copy-move, splicing, object removal, and morphing, making it well-suited for developing robust and generalizable models [4][11][16].

Each manipulated image in the dataset is accompanied by a corresponding ground-truth mask that precisely highlights the tampered regions, providing essential pixel-level supervision for segmentation models like U-Net or DeepLabV3+ [6][8][10]. These masks improve the model's ability to learn localized manipulation patterns and enhance interpretability. The dataset also includes authentic (unaltered) images, ensuring balanced and unbiased training of the classification network [16].

4.1 Dataset Preparation

Before training, the dataset will undergo the following preprocessing steps:

- **Resizing:** All images and masks will be resized to a uniform dimension (e.g., 224×224 pixels) to ensure compatibility with the network input layers.
- **Normalization:** Pixel values will be scaled to the [0,1] range or standardized for stable model convergence.
- **Augmentation:** Techniques such as random rotation, flipping, and brightness variation will be applied to improve model generalization and prevent overfitting.
- **Splitting:** The dataset will be divided into 70% training, 15% validation, and 15% testing sets to ensure reliable evaluation and reproducibility.

4.2 Previous Benchmark Results

Several benchmark studies have evaluated the performance of deep learning models on the DEFACTO and related image manipulation datasets. These benchmarks provide a useful reference for setting performance expectations for this project.

- Bayar and Stamm (2016) [6] introduced one of the earliest CNN architectures specifically designed for detecting image manipulation. Their model achieved up to 97% classification accuracy on datasets containing splicing and copy-move forgeries. However, the method was limited to image-level classification and could not localize manipulated regions.

- Ronneberger et al. (2015) [7] proposed the U-Net architecture, which has since become a cornerstone for segmentation tasks. When adapted for image forgery localization, U-Net models typically achieve Intersection over Union (IoU) scores between 0.65–0.80 and Dice Coefficients of around 0.70–0.85, depending on dataset complexity and augmentation techniques.
- Chen et al. (2018) [8] improved upon U-Net with DeepLabV3+, introducing atrous convolution and an encoder-decoder structure that enhanced boundary precision. On manipulation datasets such as CASIA and DEFACTO, DeepLabV3+ has reported IoU values exceeding 0.80 and superior segmentation of fine-grained tampered regions compared to U-Net.
- Bappy et al. (2019) [13] proposed a hybrid LSTM and encoder-decoder approach, demonstrating F1-scores above 90% for binary classification and notable improvements in spatial localization. Their work highlights the advantage of combining spatial and sequential feature learning for manipulation detection.
- Korus and Huang (2017, 2019) [15][16], in developing the DEFACTO dataset, reported that baseline CNN models achieved classification accuracies of 92–95% and segmentation IoU scores around 0.70, establishing a strong benchmark for future models. However, they also noted that model generalization across manipulation types remains a key challenge, motivating the need for multi-architecture and hybrid frameworks.

4.3 Rationale for Dataset Selection

The DEFACTO dataset was chosen for its variety of manipulation techniques, availability of ground-truth masks, and realistic image quality, which collectively make it a suitable choice for a deep learning model aimed at both forgery detection and localization. Its diversity across manipulation categories enables the proposed system to generalize effectively to unseen forgeries, thereby improving real-world applicability in digital forensics and image verification tasks.

5.0 Proposed Work and its Methodology

The project will follow a structured methodology with clearly defined steps:

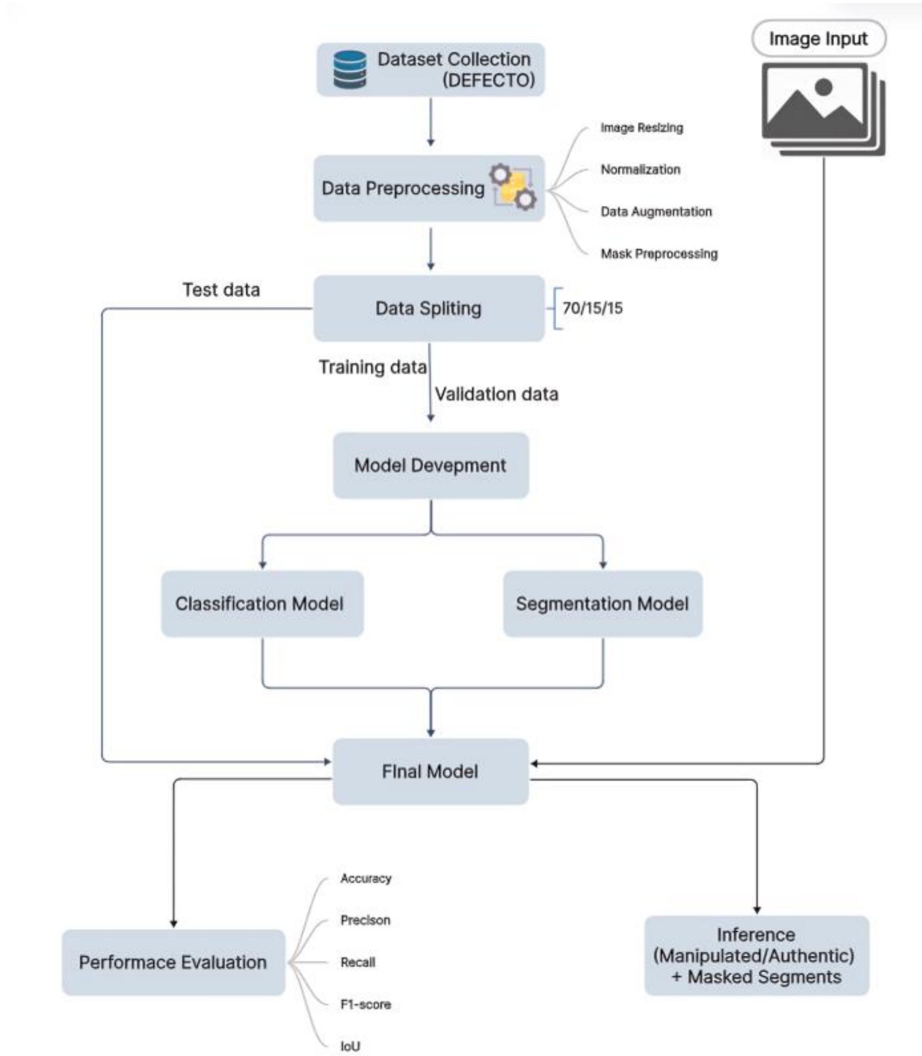

Figure 1: Abstract Model of the Proposed Image manipulation detection and segmentation

This figure proposes a deep learning-based approach for detecting and localizing manipulated regions in images using the DEFACTO dataset. The overall workflow consists of dataset collection, preprocessing, data splitting, model development, and performance evaluation. The figure above shows the abstract model of the proposed methodology.

5.1 Model Overview

The main goal of this project is to build a system that can both detect whether an image is manipulated and localize the manipulated areas. The proposed approach includes two major models:

1. Classification Model – determines if an image is authentic or manipulated.
2. Segmentation Model – identifies and highlights the exact manipulated regions.

Both models are trained using the pre-processed data from the DEFACTO dataset. The complete process starts with dataset preparation and ends with performance evaluation using suitable metrics.

5.2 Architecture Description

The project uses two deep learning architectures for different tasks:

- **Classification Model:** A Convolutional Neural Network (CNN) or a pretrained model like ResNet50 will be used. This model will take an image as input and output whether it is authentic or forged. CNN extracts spatial features and patterns that differentiate real images from manipulated ones.
- **Segmentation Model:** A U-Net or DeepLabV3+ model will be used for localization. It will take a forged image as input and output a binary mask showing manipulated areas. The encoder part of the network captures high-level features, while the decoder reconstructs a pixel-level mask to highlight forged regions.

This two-stage design allows the system to first detect manipulated images and then precisely locate the forged parts.

5.3 Training Strategy

After preprocessing, the dataset will be divided into training (70%), validation (15%), and testing (15%) subsets.

- **Data Preprocessing:**
  All images will be resized to a fixed dimension (e.g., 224×224), normalized, and augmented using random flips, rotations, and brightness changes. For segmentation, corresponding masks will be binarized and resized using nearest-neighbour interpolation.
- **Model Training:**
  Both models will be trained using the Adam optimizer and Binary Cross-Entropy (BCE) loss function. For segmentation, an additional Dice Loss will be combined with BCE to improve mask accuracy. Early stopping and learning rate scheduling will be used to avoid overfitting.
- **Implementation Tools:**
  The models will be implemented using TensorFlow/Keras or PyTorch, and training will be performed on GPU using Google Colab or Kaggle environments.

5.4 Performance Evaluation Metrics

The performance of the models will be measured using the following metrics:

For Classification Model:

- Accuracy: Measures overall correctness of predictions.
- Precision: Indicates how many predicted forgeries are actually forged.
- Recall: Measures how many real forgeries were correctly detected.
- F1-Score: Harmonic mean of precision and recall for balanced evaluation.

For Segmentation Model:

- Intersection over Union (IoU): Measures the overlap between predicted and actual forged regions.
- Dice Coefficient: Evaluates the similarity between predicted and true masks.
- Pixel Accuracy: Calculates the percentage of correctly labelled pixels.

These metrics will ensure that both detection and localization performance are evaluated thoroughly.

6.0 Project Timeline

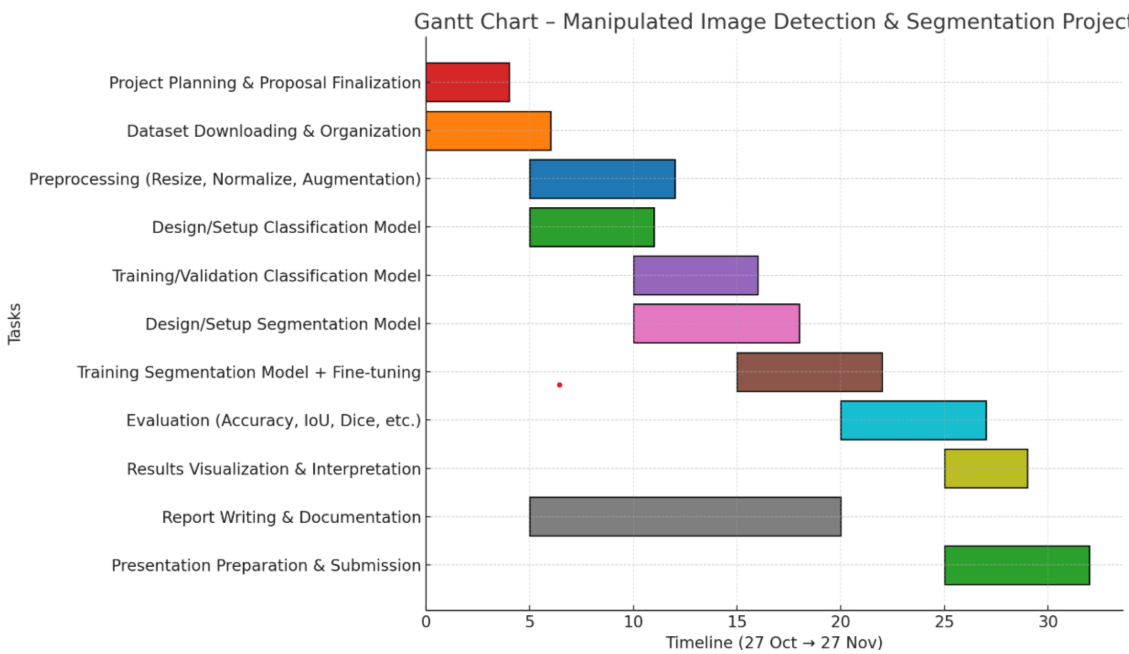

Figure 2: Gantt Chart

7.0 Conclusion

This project proposes the development of a robust deep learning system for Image Forgery Detection and Localization using a two-stage approach on the DEFACTO dataset.

The project addresses the critical need to restore trust in digital media by providing a comprehensive solution that can first classify an image as Authentic or Forged (using a CNN) and then precisely localize the manipulated pixels (using a U-Net or DeepLabV3+ segmentation network).

By employing rigorous evaluation metrics such as F1-Score, IoU, and Dice Coefficient, and targeting a high performance, the expected outcome is a highly accurate and practical tool for digital forensics. The successful completion of this project will contribute a valuable, dual-purpose model to the fight against visual disinformation and media manipulation

8.0 References

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Presentation Video Link:

https://drive.google.com/drive/folders/1njshbdvcHcRpmhQLb-zdWJAp3_lrQmH6?usp=sharing